



Global dematerialization, the renaissance of Artificial Intelligence, and the global stakeholder capitalism model of digital platforms: current challenges and future directions

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Abstract

In this article, we explain how the data-driven approach to Artificial Intelligence enabled by global dematerialization and embraced by digital platforms leads to an “excess of unutilized big data” and why the need to unlock the value hidden in increasingly larger volumes of big data leads to a new model of how digital platforms drive and profit from innovation that departs from the standard model. We also explain how this new model is based on a dual positioning of big data as a core and as a generalized complementary asset in the “extended innovation ecosystems” of digital platforms and how the rise of these extended innovation ecosystems ultimately leads to the rise of the “global stakeholder capitalism model of digital platforms” as the new economic model under which digital platforms operate today. We conclude with a discussion of some of the challenges and future directions of this new model at the intersection of Artificial Intelligence, innovation and societal transformation.

Keywords Digital platforms · Global dematerialization · Big data · Extended innovation ecosystems · The global stakeholder capitalism model of digital platforms

JEL Classification O30 · O31 · O32 · O33 · O36

1 Introduction

Digital platforms operate under an economic model sharing some of the elements of the model of global hypercompetition (D’Aveni 1994) and the global stakeholder

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capitalism model (Nolan 2008; Freeman et al. 2007). A crucial distinction of this new model is that it leads to strong global network effects that can cause digital platforms to take all or almost all the market very rapidly. Keeping the Internet unregulated is a necessary condition for digital platforms to expand globally and is thus a key political assumption of this new model.

Contrary to conventional pipeline businesses that operate in linear value chains (Parker et al. 2016; van Alstyne et al. 2016), digital platforms spark value creation processes that draw upon demand-side economies of scale, which are in turn fueled by strong network effects in value creation systems (Normann 2001). But these same network effects can also spark processes of value destruction that can erode the business model of global digital platforms in much the same way. Even in the absence of competition, the markets global digital platforms serve remain contested markets. In response to this contingent threat, digital platforms need to innovate constantly under a form of hypercompetition that goes beyond the one described by D'Aveni (1994). In fact, the inherent fragility of their business models forces digital platforms to innovate constantly, even in the absence of direct competitors.

1.1 The role of Artificial Intelligence

While digital platforms triggered the renaissance of Artificial Intelligence during the first two decades of the new millennium, Artificial Intelligence is now igniting the global expansion and consolidation of digital platforms and is expected to continue to do so in the future. But the role of Artificial Intelligence is much more profound and is giving rise to a new economic model that has been referred to as the “global stakeholder capitalism model of digital platforms” (Paredes-Frigolett and Pyka 2022). Although it borrows some of the characteristics of the global hypercompetition model (D'Aveni 1994) and the global stakeholder capitalism model (Hsieh 2004; Freeman et al. 2007; Nolan 2008), the global stakeholder capitalism model of digital platforms is new and is currently disrupting industries dominated by incumbents with conventional pipeline business models of the old economy (van Alstyne et al. 2016).

1.2 How digital platforms drive innovation and profit from it

Big data represents an unprecedented opportunity for digital platforms. But in order to capitalize on it, digital platforms need to explore and discover new valuable uses for increasingly larger volumes of big data and implement new business models to monetize them. As a general purpose technology platform, Artificial Intelligence holds the key to unlocking and harnessing the value of big data. But the application of Artificial Intelligence to solve this problem is a tough nut to crack that requires the development of cospecialized assets in a potentially very large space of potential application domains under radical uncertainty (Kay and King 2020).

The key research question addressed in this article is therefore whether this process of exploration and discovery shall be conducted in stand-alone mode by letting digital platforms position big data as a core asset that is kept captive to their own services, or whether they should extend their innovation ecosystems by positioning big data as a

generalized complementary asset that can be used by a potentially very large number of stakeholders in order to achieve a superpositioning effect.¹

1.3 Research question

The research question mentioned above can be stated using the standard model of how firms profit from innovation (Teece 1986, 2006). More specifically, the research question consists in whether or not the following hypothesis holds true in the case of digital platforms.

1.3.1 The internalization hypothesis

Applied to digital platforms, the internalization hypothesis can be stated as follows: “if the digital platform does not possess the complementary assets that are needed to unlock the value of its big data and profit from it, if these assets are critical and specialized, if the digital platform can develop them internally, and if competitors or imitators are not better positioned to develop them, then the digital platform should internalize those assets” (Teece 1986, p. 296).

1.3.2 The unit of analysis

The unit of analysis we will use is the big data digital platforms compile by facilitating very large volumes of human-machine interactions around the so-called value unit of digital platforms. Defined as the unit around which the interactions facilitated by the digital platforms revolve (Parker et al. 2016), the value unit is an essential component in the initial design phase of any digital platform, which continues to play a key role in harvesting large volumes of data during the subsequent phases of the lifecycle of digital platforms. Examples of such value units are professional profiles in the case of LinkedIn and online videos in the case of YouTube.

1.3.3 Driving and profiting from innovation

The internalization hypothesis, as stated above, corresponds to how firms that operate under weak appropriability regimes drive and profit from innovation according to the standard model (Teece 1986, 2006, 2018a, b). Big data, which results from billions of interactions facilitated by digital platforms around the value unit, needs to be monetized by finding new valuable uses. In order to drive innovation and profit from it, digital platforms need to find such valuable uses, define revenue models to monetize them, and implement business models to deliver new value in their innovation ecosystems. As digital platforms operate under weak appropriability regimes, the standard model

¹ Such a superpositioning effect is one of three quantum-mechanical effects mimicked in quantum management, a novel strategic management framework inspired by quantum mechanics (Paredes-Frigolett 2023). The superpositioning effect is achieved by letting stakeholders in the extended innovation ecosystem of digital platforms explore, discover and implement multiple innovation pathways in parallel, thus setting off value creation processes in a burst of activity both in time and space.

would suggest that digital platforms should pursue a strategy of internalization of the complementary assets needed to drive innovation and profit from it.

1.3.4 Testing existing theories through case research

Case-based research can serve the purpose of theory building in areas in which theory does not yet exist or theory testing and elaboration in case theories already exist (Ketoviki and Choi 2014). We will take the standard model as the general theory (Teece 1986, 2006, 2018a, b). As the standard model allows us to formulate our research question and derive the explicit internalization hypothesis stated above (Ketoviki and Choi 2014, p. 238), we will test whether the internalization hypothesis holds true in the case of digital platforms.

1.4 Organization of the article

The article is organized as follows. In Section 2, we describe previous work our research builds upon. In Section 3, we introduce several cases that show how digital platforms drive innovation and analyze whether or not the way they profit from it departs from the standard model (Teece 1986, 2006, 2018a, b). In Section 4, we introduce a new model of how digital platforms drive and profit from innovation and we discuss: (i) how global dematerialization, as the process underlying the generation of big data, leads digital platforms to position big data both as a core and as a generalized complementary asset; (ii) why this dual role of big data leads to a new model of how digital platforms drive and profit from innovation; (iii) why this new model is based on extending the innovation ecosystems of digital platforms; and (iv) why the resulting extended innovation ecosystems lead to the rise of the global stakeholder capitalism model of digital platforms and to the positioning of stakeholder engagement as the source of competitive advantage of digital platforms. In Section 5, we conclude with a discussion of current challenges and future directions.

2 Previous work

2.1 Defining digital platforms

As noted by Gawer and Cusumano (2014, p. 418), and more recently by de Reuver et al. (2018, p. 125), platforms have been usually grouped in three main categories.

2.1.1 Internal platforms

Gawer and Cusumano (2014, p. 418) define internal platforms as “a set of assets organized in a common structure from which a company can efficiently develop and produce a stream of derivative products.” As these authors point out, the innovations in internal platforms are driven within the firm. These platforms existed long before Internet technologies emerged in the mid 1990s and are not the type of platforms we will deal with in this article.

2.1.2 Supply chain platforms as a form of internal platforms

A special case of internal platforms are the so-called “supply chain platforms” (Gawer and Cusumano 2014, p. 419). These platforms operate in complex linear value chains and also preceded Internet technologies. Supply chain platforms are also outside of the scope of digital platforms, as we define them in this article.

2.1.3 External platforms

Gawer and Cusumano (2014, p. 420) define external platforms² as “products, services, or technologies developed by one or more firms, and which serve as foundations upon which a larger number of firms can build further complementary innovations and potentially generate network effects.” These external platforms are different from internal digital platforms in that the value creation process in the case of external platforms is open to other firms. This leads to the concept of open innovation systems comprising other firms that act as complementors. Although external platforms are predicated on open innovation as a feature that is shared by digital platforms, they predated Internet technologies and do not include key defining features of digital platforms.

2.2 Our definition of digital platforms

Our definition of digital platforms is based on the distinction between pipeline and platform business models proposed by Parker et al. (2016). Pipeline business models operate in linear value chains and focus on value creation through products and services. Digital platforms, on the other hand, create value not in linear value chains but in complex value creation systems and focus on facilitating interactions.

We define digital platforms not from a technology management or information systems perspective (Hein et al. 2019, p. 88) but rather from a market-based perspective (Parker et al. 2016). Below we summarize the key defining features of digital platforms, as described by Paredes-Frigolett and Pyka (2022):

1. digital platforms use the Internet as a network infrastructure commons but not all business models based on the Internet, usually referred to as e-businesses (Amit and Zott 2001), are digital platforms;
2. digital platforms create value not in linear value chains but in value creation systems, which correspond to complex innovation ecosystems that digital platforms build, extend, and orchestrate (Parker et al. 2016);
3. as opposed to internal and external platforms, digital platforms are not in the business of developing products and providing value-added services based on those products but rather on facilitating key interactions through the platform and providing value-added services based on those key interactions;
4. the competitive advantage of digital platforms is neither predicated on deploying the largest network infrastructure nor on providing the best content but rather on providing the best facilitation (of key interactions);

² Also called industry platforms.

5. as opposed to e-businesses that use the Internet as delivery model and drive network effects based on Metcalfe's law (Hendler and Golbeck 2008) but still operate under the logic of a legacy (pipeline) business model (Parker et al. 2016), digital platforms create value jointly with complementors³ and drive much stronger network effects in group-forming networks (Reed 2001);
6. as digital platforms have the potential to create new markets and industries from scratch, they can exploit unknown limitations of market-based systems in unregulated industries, which creates the need for more complex governance models than those used by internal and external platforms;
7. using such governance models, digital platforms have to regulate price and competition and they also have to act as licensing authorities and enforcers⁴ (Rochet and Tirole 2003, 2006) and as policymakers in the areas of monetary policy, industrial policy, IP policy and foreign policy⁵ (Parker et al. 2016).

2.3 The global stakeholder capitalism model of digital platforms

Hein et al. (2019) have recently proposed a characterization of different digital platform ecosystems based on their ownership structure, their value creation mechanisms, and the autonomy of their complementors.

In our model, we assume that the digital platform is not owned by a consortium or a peer-to-peer community but by a single owner (Hein et al. 2019, p. 90).

While the sides that are part of the innovation ecosystem contribute to the value creation mechanisms of the platform through data generated unobtrusively and on the fly as they interact with the platform, they also contribute to innovations, especially in the form of user-generated content and services. The interested parties and their stakeholders in the extended innovation ecosystem contribute to value creation mostly through innovation. They play an active role in diversifying the range of services offered by engaging in the exploration, discovery, and exploitation of new services in the extended innovation ecosystem (Hein et al. 2019, p. 91).

While the sides in the innovation ecosystem tend to be more tightly coupled through the governance model of digital platforms, the interested parties and stakeholders in the extended innovation ecosystem are more loosely coupled and more autonomous (Hein et al. 2019, p. 92).

The global stakeholder capitalism model of digital platforms is shown in Fig. 1 below.⁶

³ Complementors can have different degrees of autonomy (Hein et al. 2019, p. 92).

⁴ See <https://www.nber.org/lecture/keynote-address-jean-tirole-future-platform-regulation> for a discussion about how digital platforms can act as regulators.

⁵ See <https://www.usv.com/writing/2010/06/web-services-as-governments/> for a discussion about the analogy between digital platforms and governments proposed by Brad Burnham from Union Square Ventures.

⁶ We reproduce Figure 1 from (Paredes-Frigiolett and Pyka 2022, p. 477).

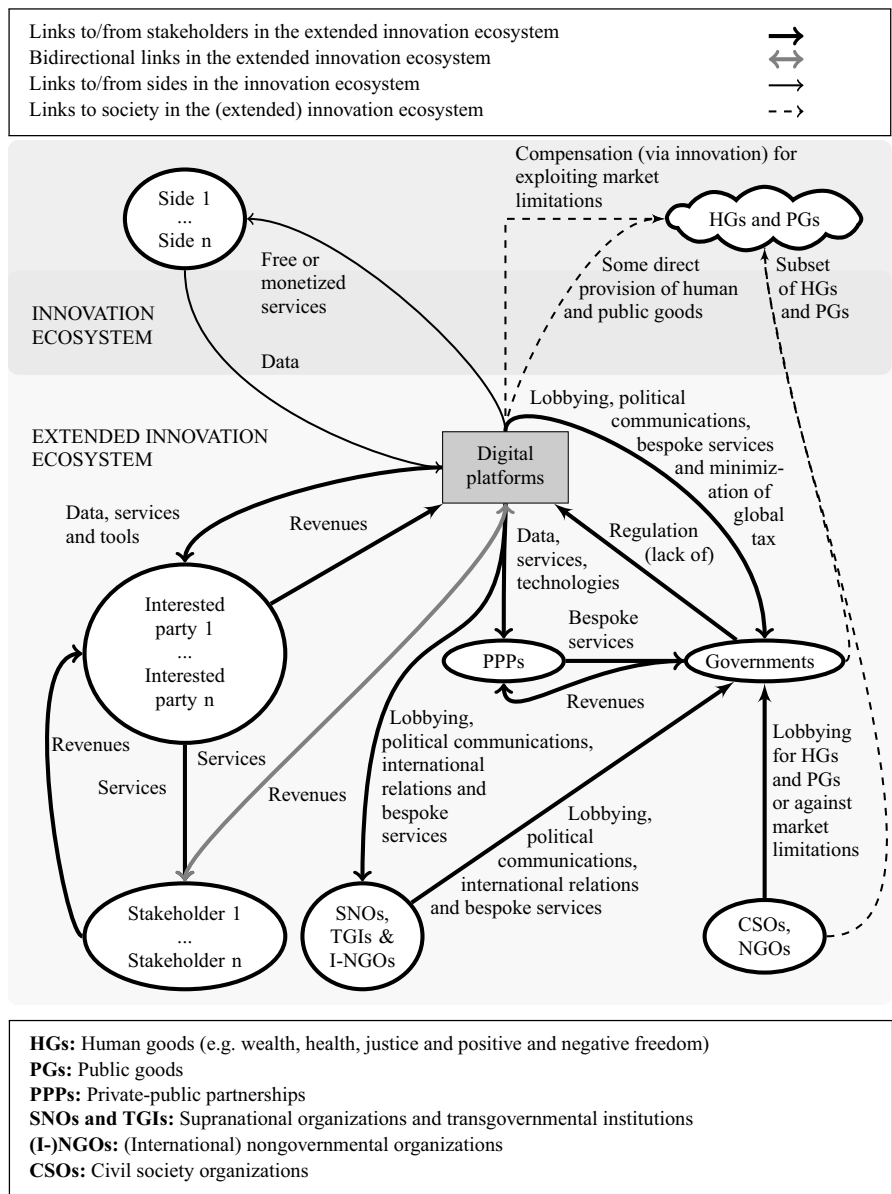


Fig. 1 The global stakeholder capitalism model of digital platforms according to Paredes-Frigolett and Pyka (2022, p. 477)

Table 1 below shows the characteristics of the global stakeholder capitalism model of digital platforms using the activities needed to solve the general strategic problem of the firm, as introduced by Singer (2013, p. 217-219), including cooperating with

Table 1 Characteristics of the global stakeholder capitalism model of digital platforms based on the model set out by Paredes-Frigiolett and Pyka (2022)

Activity	Characteristics
Strategy	“Coopeting” strategies in “extended innovation ecosystems” and the “extended governance of these ecosystems,” which are based on global dematerialization as the key political assumption of the global stakeholder capitalism model of digital platforms, are the source of competitive advantage
CSR	Deliver public and human goods globally by diffusing innovation spillovers as the central pillar of CSR strategy
CSOs/NGOs I-NGOs	Partner with CSOs/NGOs/I-NGOs promoting corporate negative freedoms and the regulatory (antitrust) status quo
PCs and lobbying	Promote corporate negative freedoms; lobby to avoid/preempt digital (antitrust) legislation and regulation and exploit global opportunities by providing bespoke services through PPPs
Legislation/regulation	Keep regulatory status quo through self-regulation, self-restrain, preemption, and compensation by delivering human and public goods
Market limitations	Exploit known and future limitations of digital platform markets in spirit of global “coopetition” compensating either for narrow interests (e.g. for reputational effects) or with a genuine dual mission of serving society and all stakeholders as ends in themselves
Taxation	Minimize tax globally and negotiate bespoke tax regimes.
SNOs and TGIs/Governments	Partner with SNOs and TGIs and governments to expand corporate negative freedoms, consolidate global data-industrial complexes, and secure a future role as an instrument in global geostrategy and geopolitics

CSOs, NGOs and I-NGOs,⁷ driving CSR strategy,⁸ engaging in PCs⁹ and lobbying, providing bespoke services through PPPs,¹⁰ keeping the regulatory status quo through compensation,¹¹ and partnering with governments, SNOs and TGIs.¹²

2.4 Global dematerialization

The success of user modeling and machine learning technologies in developing commercial-grade applications in a variety of domains requires very large repositories of data (Sowa 2000). Today, these very large datasets are harvested mostly by incumbent digital platforms and are referred to as big data. To understand the impor-

⁷ Civil society organizations and (international) nongovernmental organizations.

⁸ Corporate social responsibility.

⁹ Political communications.

¹⁰ Public-Private Partnerships.

¹¹ Such as the compensation via bespoke tax agreements of Airbnb agreed upon with municipalities (Nieuwland and van Malik 2020).

¹² Supranational organizations and transgovernmental institutions.

tance of big data as a core asset of digital platforms one needs to go back to the first revolution of Artificial Intelligence, a period that is reminiscent of today's frenzy about Artificial Intelligence. This first revolution was based on the so-called symbolic paradigm of Artificial Intelligence (Chomsky 1968; Fodor 1975).

2.4.1 The long night of Artificial Intelligence

The symbolic paradigm gave rise to the field of knowledge representation as the central research area of Artificial Intelligence (Levesque and Brachmann 1987). This paradigm failed to deliver commercial-grade applications due to the theory bottleneck problem of Artificial Intelligence (Lenat et al. 1985; Wagner 2006), which is the problem of building large knowledge bases to represent the knowledge needed to enable reasoning as the core functionality of any knowledge-based system (Lenat and Feigenbaum 1987). With the exception of narrowly defined application domains, the symbolic paradigm failed to deliver commercial-grade knowledge-based systems due to the need to build and also maintain very large knowledge bases containing the world and domain-specific knowledge that was needed to enable automatic reasoning.

There were two main impediments that prevented such large knowledge bases to be built (Lenat and Guha 1990, 1993). The first problem was that the knowledge needed to be entered by knowledge engineers in a top-down fashion.¹³ The second, and more fundamental and difficult problem to solve, was the computational complexity of the reasoning algorithms of mainstream knowledge representation languages, which had adopted first-order predicate logic as the logical foundation for automated reasoning in Artificial Intelligence (Nebel and Smolka 1991; Smolka 1992). To avoid this second problem, mainstream knowledge representation research focused on the development of logical frameworks based on subsets of first-order predicate logic, whose reasoning algorithms, although more computationally tractable, did not have the “expressiveness” to represent the world and domain-specific knowledge needed for the development of commercial-grade knowledge-based systems (Gabbay et al. 1993). This gave rise to the famous trade-off of expressiveness of knowledge representation, which divided the research community throughout the late 1980s and 1990s into those who proposed the full power of first-order predicate logic and those who proposed restricted subsets of this logic to implement commercial-grade knowledge-based systems.

After four decades of research and development since the 1950s, the symbolic approach proved unable to deliver on its promise and led to what went down in history as “the long night of Artificial Intelligence,” a research-funding hiatus that unfolded through the 1990s and continued on after the turn of the millennium.

2.4.2 Back from the dead

Based upon a data-driven, as opposed to the knowledge-based approach, a new sub-symbolic paradigm that was inspired by the neurosciences gained traction and began

¹³ Although with the emergence of the Internet some approaches were proposed to accomplish this task automatically (van Durme and Schubert 2008; van Durme et al. 2009, 2008).

to deliver better results due to the sudden availability of very large datasets in a variety of application domains (Churchland and Sejnowski 1992). But this new paradigm of Artificial Intelligence, with its new potential to deliver commercial-grade systems, could only take off after the rise of the digital platform revolution (Parker et al. 2016) for the reasons we analyze below.

The convergence of several factors put an end to the long night of Artificial Intelligence. The first one was the rise of the data-driven subsymbolic paradigm of Artificial Intelligence, which replaced the knowledge-based symbolic paradigm and became the new mainstream approach to Artificial Intelligence at the turn of the millennium. Yet despite all its bravado, the new subsymbolic approach suffered from the same problem as the previous symbolic approach, namely, it needed the availability of large datasets, instead of large knowledge bases, to feed machine learning algorithms and render them useful. Its main advantage, however, was that these datasets were generated in a bottom-up instead of a top-down fashion as a result of digital platforms facilitating human-computer interactions (Rich 1979; Kobsa 2001). Not requiring the intervention of big battalions of knowledge engineers, the process of harvesting large datasets could unfold automatically.

Despite this enormous advantage, it was not until digital platforms consolidated after the first decade of the new millennium that the sudden explosion of big data sparked the renaissance of Artificial Intelligence. Today, Artificial Intelligence based on the data-driven subsymbolic paradigm and fueled by big data is driving the innovation strategy of digital platforms (Sivinski et al. 2017). This begs the question of what process led to such explosion of big data in the first place.

2.4.3 Global dematerialization and the rise of digital platforms

The subsymbolic paradigm of Artificial Intelligence benefits from the sheer volume of human-computer interactions that are volunteered on a daily basis by the “sides” of multisided digital platforms (van Alstyne et al. 2016). The mediating factor for the rise of big data is a process that takes place today on an unprecedented large and global scale. This process, which has been referred to as dematerialization (Paredes-Frigolett and Pyka 2022),¹⁴ can be defined as the process used by digital platforms to:

1. build digital representations of individuals, processes, and things that could be located anywhere in the world unobtrusively and on the fly;¹⁵
2. augment these digital representations using machine learning and user modeling algorithms driven by big data, thereby building into them global “wisdom-of-the-crowds” and “power-of-us” effects;¹⁶

¹⁴ Digital dematerialization, or simply dematerialization, is a term originally suggested by Eric K. Clemons.

¹⁵ The representations are technically known as (user) models.

¹⁶ “Wisdom of the crowds” and “power-of-us-effects” are implemented through machine learning techniques such as collaborative filtering, which is used to instantiate recommendations for a given user based on her/his user model, on the one hand, and a very large number of similar user models, on the other. This process, referred to as “user modeling,” occurs unobtrusively and on the fly and uses the power of millions of user models for personalization and recommendation purposes. Intra- and inter-session user modeling leads to the augmentation of user models as digital representations, which in turn leads to these “wisdom-of-the-crowds” and “power-of-us” effects.

3. deploy these “augmented” digital representations so as to unbundle, rebundle, configure and reconfigure personalized value offerings in real time; and
4. deploy these offerings to “audiences of one” anytime and anywhere.¹⁷

The last point is very important because if digital platforms are allowed to expand and operate globally in a seemingly frictionless way, they can use the process of dematerialization to deploy offerings in innovation ecosystems that operate globally. Dematerialization becomes then “global dematerialization.” Dematerialization is the process that leads to the generation of big data as a core asset of digital platforms and is of paramount strategic importance not only for digital platforms that dominate globally but also for those that dominate in regional and niche markets.¹⁸

2.5 The rise of the extended innovation ecosystems of digital platforms

Digital platforms were instrumental in the renaissance of Artificial Intelligence during the first decade of the 21st century through the global dematerialization process described above. The exponential growth of big data driven by digital platforms has created an excess capacity of large volumes of unutilized big data. This, in turn, has given rise to a new strategic problem that is of the utmost importance for digital platforms going forward, namely, “the need to unlock the value hidden in large volumes of unutilized big data.”

Global dematerialization allows digital platforms to compile and augment their digital representations by generating datasets that are more valuable not only in terms of coverage, that is, access to a larger number of users in different markets, but also in terms of granularity, that is, the number of attribute-value pairs they can infer via inter- and intra-session user modeling (Rich 1979; Kobsa 2001). These two properties are of strategic importance not only to consolidate the significant market power of global digital platforms but also to prevent other digital platforms from emerging in peripheral markets and competing with them.

Global dematerialization is of strategic importance for digital platforms because it allows them not only to dramatically increase the density of services,¹⁹ defined as the number of service offerings catering to “audiences of one,”²⁰ but also to deploy such offerings globally drawing upon “wisdom-of-the-crowds” and “power-of-us effects”²¹ fueled by big data generated anytime and anywhere.

2.5.1 Big data as a core asset of digital platforms

Big data is a core asset of digital platforms that has no direct monetizable value. To unlock its value, the following process has to take place:

¹⁷ A condition for which digital platforms need to be allowed to operate globally.

¹⁸ As shown by KakaoTalk and its expansion into North and South East Asia.

¹⁹ A term coined by Normann (2001).

²⁰ These audiences correspond to segments comprised of one consumer and are usually associated with digital platforms that deploy user personalization algorithms such as collaborative filtering (Schafer et al. 2007).

²¹ Which we introduced in Section 2.4.3.

1. big data needs to be converted into information by assigning a meaning to it;
2. new uses need to be found for this information, thereby transforming information into new knowledge, which is a process that entails the exploration and discovery of how new information can solve new important needs of potential clients or users and lead to value creation; and
3. a viable revenue model for value capture needs to be explored, discovered and implemented in order for the value created to be monetized.

The first two steps of this three-step process are well-known in information sciences (Devlin 1999) and deal with the problem of value creation. However, the third step deals with the problem of value capture, which is a critical step in order to monetize the value discovered in the first two steps that can prove elusive. As the amount of unutilized big data explodes, this three-step process becomes computationally intractable due to the high strategic uncertainty associated with exploring and discovering new valuable uses in an increasingly larger search space. This begs the question of whether dominant digital platforms should engage interested parties and stakeholders to unlock the value hidden in unutilized big data and profit from it.

2.5.2 Big data as a core or as a complementary asset of digital platforms?

The key to extending the innovation ecosystem of digital platforms with interested parties and their stakeholders is the positioning of big data as a generalized complementary asset (Teece 1986, 2006; Paredes-Frigolett and Pyka 2022). This strategy might seem controversial at first glance. Indeed, the standard model would suggest that big data should be used to deploy future services by the platform alone (Teece 1986, 2006, 2018a, b). But while such a strategy may be successful when the innovations of firms are not subject to deep strategic uncertainty and when value capture is governed by strong as opposed to weak appropriability regimes, empirical evidence suggests that the pursuit of this strategy often constitutes a strategic folly for digital platforms (Yoffie et al. 2019).

2.6 The extension of the innovation ecosystem and its dilemma

As global dematerialization becomes ubiquitous, digital platforms are left with the problem of how to unlock the value hidden in increasingly larger volumes of big data, which involves a three-step process that is ultimately aimed at monetizing new uses for big data. As introduced by Paredes-Frigolett and Pyka (2022), the dilemma of innovation ecosystem extension digital platforms face consists in whether to provide their big data as a generalized complementary asset to interested parties and stakeholders in order to let them use it to explore, develop and deploy the cospecialized assets needed to find and monetize new valuable uses or whether to internalize this process by keeping big data captive to their own future service offerings. This dilemma is of strategic importance for digital platforms and is difficult to solve for it involves a trade-off between increasing flexibility to find and deploy the cospecialized assets needed to boost value creation in the extended innovation ecosystem of digital platforms, on the one hand, and securing command and control of the cospecialized assets

Table 2 Case selection criteria

Criterion	Characteristics
Launch	The digital platform has already undergone the launch phase successfully
Expansion	The digital platform has already expanded regionally or globally
Consolidation	The digital platform has consolidated as incumbent digital platform
Monetization	The digital platform has already found a viable revenue model
Extension	The digital platform has already initiated a process of expansion beyond the sides originally served by the platform.
Grand strategies	The digital platform is already executing a grand strategy (such as the new grand strategies of Alphabet and Meta Platforms)

to boost value capture for the platform, on the other. According to the internalization hypothesis introduced in Section 1, digital platforms should pursue the latter strategy to deal with this dilemma.

In the next, section, we analyze several cases showing how digital platform did address the dilemma of innovation ecosystem extension.

3 Dealing with the dilemma of innovation ecosystem extension

In order to shed light on the research question presented in Section 1.3 we analyze cases of digital platforms according to the following criteria.

3.1 Case selection criteria

The cases we use to validate the internalization hypothesis consist mostly of incumbent digital platforms with global presence. Accordingly, we will analyze in this section cases of digital platforms satisfying the criteria shown in Table 2.²²

3.2 Cases of innovation ecosystem extension

3.2.1 The case of MySpace versus Facebook

One of the first cases of how to address the dilemma of innovation ecosystem extension is the case of MySpace after the turn of the millennium, a period in which MySpace decided to keep its core assets, including big data, captive to their own future service

²² We may drop some of these criteria to include cases of incumbent digital platforms that have either not yet initiated the execution of grand strategies or have not yet found a viable revenue model.

offerings. Acquired by News Corporation in 2005, the strategy of MySpace shifted from value creation via open innovation to value capture. This radical shift was in accordance with the standard model (Teece 1986, 2006) and gave Facebook a window of opportunity to execute an opposite strategy of innovation ecosystem extension. By 2009, Facebook had overtaken MySpace's place as the world's leading social media platform in terms of active users (Jacobs 2015). This raises the question of what went wrong with MySpace's strategy.

After this acquisition in 2005, the new management team of MySpace promptly began to address the innovation ecosystem extension dilemma differently by executing a strategy of internalizing the cospecialized assets needed to innovate. MySpace's new strategy aimed at maximizing value capture and consisted in finding new valuable uses for its big data without recourse to a heterogeneous group of interested parties and stakeholders. In so doing, MySpace fully embraced the internalization hypothesis. As this new strategy did not aim at maximizing value creation for the extended innovation ecosystem, MySpace's large user base began to perceive less value in MySpace than in other emerging competitors such as Facebook. As a result, the generation of demand-side economies of scale MySpace could capitalize upon was seriously compromised, which allowed Facebook to catch up and eventually surpass MySpace in terms of active users.

3.2.2 How Google ignited its first period of hypergrowth

Syndicating digital content on the Internet had been around in the form of Web syndication since the mid 1990s, when Apple implemented the Meta Content Framework within its Advanced Technology Group (Guha 1996). But Google's most important innovation before going public was the syndication of a very special type of digital content, namely, online ads.

Google's idea of syndicating online ads was a revolutionary development that had a profound impact on the firm. Google created the online ad syndication market after the turn of the millennium by syndicating ads with any website based on keywords using the same CPC²³ revenue model that Google had successfully deployed to sell computational advertising on its own site (Werbach 2000). Instead of keeping them captive to its future offerings, Google positioned its keywords as a complementary asset in a new extended innovation ecosystem. Through the AdWords and AdSense programs launched before its IPO, Google positioned millions of businesses with an online presence as online publishers by offering them revenue-sharing agreements as online syndication partners. The original innovation ecosystem of Google as a two-sided platform was thus extended with millions of small and medium-sized businesses, which became online syndication partners in an extended innovation ecosystem orchestrated by Google. This strategy of innovation ecosystem extension allowed Google to capitalize upon very large volumes of online ad syndication transactions generated on the site of millions of businesses globally. The execution of this strategy through Google's

²³ Cost per click-thru.

AdWords and AdSense programs²⁴ allowed the company to enter a first period of hypergrowth that led to its IPO in 2004 (Stross 2009) and to its further consolidation as the world's largest provider of search engine advertising (Sullivan 2016).

This case shows how Google solved the innovation ecosystem extension dilemma by dropping the internalization hypothesis. Google did not keep keywords as a form of big data captive to its own offerings in an attempt to appropriate all the profits from its search engine marketing business based on keywords. By dropping this hypothesis and sharing billions of its keywords with online syndication partners, Google extended its innovation ecosystem with online syndication partners to launch a new service offering that did not require the development of cospecialized assets based on big data.

Our next case shows how extending the innovation ecosystem can have as objective the development of such cospecialized assets.

3.2.3 The case of Twitter

Twitter's extension of its innovation ecosystem aimed at implementing a revenue model for value capture (Welch and Popper 2015). In an attempt to explore several pathways simultaneously to monetize its tweets, Twitter put together an eclectic innovation ecosystem of interested parties, providing them with access to its tweets as a form of big data and letting them use it to develop a variety of technologies as cospecialized assets in areas such as predictive analytics and Internet sentiment analysis. Twitter's ultimate goal with this strategy was to find a viable revenue model to monetize its tweets. However, instead of allowing the interested parties that were successful in finding viable data businesses to profit from them, Twitter ended up internalizing the cospecialized assets developed by these interested parties in order to absorb their data businesses (Morgan 2015). As a result, Twitter earned harsh critique from interested parties for not having communicated in advance its plans to internalize the data business of interested parties and for rendering them redundant (Ingram 2011).

The case of Twitter shows how an incumbent digital platform solved the innovation ecosystem extension dilemma by implementing a strategy that rejected the internalization hypothesis. Twitter met all the conditions of this hypothesis to develop the cospecialized assets required internally. But Twitter decided to position its big data as a generalized complementary asset instead, enticing interested parties to drive new innovations. Twitter thus dropped the strategy of internalization in order to accelerate and expand the development of cospecialized assets. But once some of these interested parties were successful in finding new valuable uses in areas such as predictive analytics and sentiment analysis, and once they developed the cospecialized assets needed to deliver and monetize these new uses, Twitter decided to internalize these cospecialized assets (Ingram 2011). This amounted to embracing the internalization hypothesis in the end by absorbing the data businesses of interested parties and rendering them redundant. However, such a strategy is dangerous, as it violates best practices of digital platform governance (Cusumano and Gawer 2002) and can eventually compromise the willingness of interested parties and stakeholders to drive innovation in the extended innovation ecosystem of incumbent digital platforms.

²⁴ Google's original advertising programs catering to advertisers and publishers, respectively.

3.2.4 The failed attempt of Microsoft with its mobile operating system

The strategy of Microsoft to position Windows Mobile as the world's dominant mobile operating system fully embraced the internalization hypothesis. Indeed, Microsoft spent a decade executing a strategy based on internalizing the required complementary and cospecialized assets to profit from innovation. Loyal to the licensing model that catapulted Windows as the world's leading operating system, Microsoft licensed its mobile operating system and developed it internally with the aim of position it as the world's leading mobile operating system.

Following this strategy, Microsoft never surpassed a 4% market share as a vendor of its mobile operating system. After having executed it for over decade, Microsoft recognized in 2010 that its strategy had failed (Savov 2019) and decided to pursue a new strategy that consisted in acquiring Nokia's handset manufacturing business and becoming a mobile devices manufacturer. This second strategy once again failed to gel and was written off in 2015 (Collins 2020).

Microsoft strategy of embracing the internalization hypothesis around Windows mobile failed because its archrivals Apple and Google rejected this hypothesis and were able to harness the power of network effects that led to important demand-side economies of scale. The success story of Microsoft with its Windows operating system had been quite different in that the internalization strategy followed by Microsoft preceded the digital platform revolution and benefited from the network effects ignited by the initial alliance between IBM and Microsoft and by the emergence of IBM compatible personal computers, which adopted the Windows operating system not because of its superior functionality but because of the vast amount of third-party software that was available on Windows. The same success story could not be replicated with Windows Mobile because Apple and then Google adopted a platform business model that rejected the internalization hypothesis and benefited from larger demand-side economies of scale that drew upon the stronger network effects generated in their extended innovation ecosystems (Savov 2019).

3.2.5 The case of Google versus Apple

With the launch of the iPhone Apple entered the mobile devices and services industry as a late entrant in 2007. By 2010, and against all odds, Apple became the leader in that industry, dethroning Nokia as the leading mobile devices manufacturer. The success of Apple was based on the positioning of the iPhone not as a product but as a digital platform. While the business models of all the competitors in this industry were basically pipeline business models (Parker et al. 2016; van Alstyne et al. 2016), Apple took the entire industry by storm with a platform business as a hitherto untried business model (Yoffie et al. 2019).

All incumbents were essentially product companies in 2007. However, Apple launched the iPhone not as a product but rather as a digital platform through the Apple Store. This positioning enabled Apple to outcompete all incumbents through stronger network effects. Apple dethroned Nokia from its leading position in what can be regarded as one of the most successful catch-up strategies in the information technology industry (Alcacer et al. 2014; Cord 2014; Kenney and Pon 2011). With

92% share of profits by 2013, Apple became the most profitable handset manufacturer in the mobile devices and services industry (Häikiö 2002). In 2013, the handset manufacturing division of Nokia was acquired by Microsoft (Doz and Wilson 2017).

Apple did not fully reject the internalization hypothesis because it kept a tight control of its platform using a strategy of innovation ecosystem extension that followed a “walled garden” approach (Skilton 2016). Unlike Google, Apple did not position the iOS mobile operating system as a complementary asset. As they were direct competitors, Apple did not allow iOS to run on the devices of other manufacturers of mobile devices. Apple internalized the cospecialized technologies required to introduce incremental and substantial innovations to its mobile operating system, not granting third-party developers access to its source code. While its ecosystem did include third-party application developers, Apple retained tight control over third-party application development through its APIs²⁵ and SDKs²⁶ (McCray et al. 2011).

While the strategy of innovation ecosystem extension executed by Apple was successful in dethroning Nokia from its dominant position in the mobile devices and services industry (Vuori and Huy 2016), it would prove incapable of dealing with the attack that Google would launch with Android as its mobile operating system. Its strategy of partially rejecting the internalization hypothesis allowed Apple to secure a lion’s share of profits in the mobile devices and services industry and was successful because most of their competitors were traditional product companies with very limited potential to generate strong network effects and drive demand-side economies of scale. But Google’s launch of Android with the aim of positioning it as a leading mobile operating system would soon change the competitive landscape for Apple.

Google acquired Android in 2005 and developed it further to match the functionality of the iOS mobile operating system (Callaham 2018). Google then launched Android in 2008 as a multiple purpose platform with the goal of quickly gaining traction with mobile users. To attain this goal, Google extended its innovation ecosystem by positioning Android as an open source project, making it freely available to handset manufacturers and application developers (Yoon 2014). Google’s strategy was predicated on extending its ecosystem with two types of important stakeholders. The first type consisted of all manufacturers that competed with Apple. This first type of stakeholders was strategically important because through these manufacturers Google could get access to millions of mobile users globally. The second type consisted of thousands of application developers. This second type of stakeholders was of equal strategic importance because through developers Google would be able to ignite innovation around Android and the Google Play platform (Kenney and Pon 2011). Unlike Apple, Google extended its ecosystem by making Android available to developers and manufacturers through open APIs and by positioning it as a generalized complementary asset in its extended innovation ecosystem.²⁷ The ultimate strategic objective pursued by making this complementary asset available to these new stakeholders was

²⁵ Application programming interfaces.

²⁶ Software development kits.

²⁷ Positioning Android as an open source platform needed by manufacturers and third-party developers to launch innovations and profit from them.

to make the diffusion of Android among users frictionless (Yoon 2014). However, achieving this objective was anything but simple for Google.

As the mobile operating system controlled direct access to mobile users as the most important group of stakeholders in the extended innovation ecosystem, relinquishing ownership of and control over this asset was not an easy decision for manufacturers (Kenney and Pon 2011). For leading manufacturers that had already developed their own mobile operating systems, such as Samsung, the decision to adopt Android was even more difficult to make (Hamwi 2016).

Other competitors such as Microsoft had already entered into agreements with manufacturers to provide its mobile operating system. But Microsoft kept tight control over this asset instead. Moreover, Microsoft did not position its mobile operating system as a complementary asset. Unlike Google, Microsoft did not position Windows Mobile as an open source mobile operating system that manufacturers could use and extend freely. Loyal to its tradition, and consistent with the internalization hypothesis, Microsoft kept tight control of this core asset and failed to build a heterogeneous innovation ecosystem capable of delivering the innovations needed to catch up with Apple (Yoffie et al. 2019).

Granting key stakeholders greater control over Android's development roadmap (Boudreau 2010), Google was able to provide a much higher value proposition and succeeded in enticing several manufacturers to adopt Android as their mobile operating system. This, in turn, triggered other manufacturers to adopt the same strategy. As a result, all of Apple's competitors ended up standardizing on Android as the mobile operating system (Savov 2016). The positioning of Android as an incumbent mobile operating system in the mobile devices and services industry allowed Google to extend its innovation ecosystem even further with third-party developers, which had now strong incentives to develop mobile applications offered on Google Play (Kenney and Pon 2011). Google's success with this strategy was astonishing, as evidenced by the number of applications available on Google Play, which had already exceeded those available on the Apple Store by 2014 (Parker et al. 2016). Executing this innovation ecosystem extension strategy, Google dethroned Apple's iOS by positioning Android as the de facto mobile operating system in this industry.

Two key elements of this strategy were the decision not to compete with manufacturers of mobile devices and the creation of an extended innovation ecosystem around Android with the goal of maximizing value creation for the industry instead of value appropriation for Google (Shapiro and Varian 1999; Kenney and Pon 2011). In so doing, Google radically departed from the precepts of the Teecean model (Teece 1986, 2006). Indeed, Google rejected the internalization hypothesis by positioning its core assets as generalized complementary assets in its extended innovation ecosystem. Although still the most profitable manufacturer in this industry following its walled garden approach to innovation ecosystem extension, Apple lost its once dominant market position to Google. By dethroning Apple, Google secured not only a leading market position that grants the company direct access to billions of mobile users globally but also the capability to globally dematerialize them anytime and anywhere (Sullivan 2016). Whether Apple can continue to compete with Google with its walled-garden approach remains to be seen (Skilton 2016).

3.2.6 Google under Alphabet

The new grand strategy of Google under Alphabet shows that the extended innovation ecosystems of digital platforms will play an increasingly strategic role in the new grand strategies of incumbent digital platforms going forward. There are several reasons why the execution of this new grand strategy required a corporate restructuring. According to CBSInsights,²⁸ the founders of Google wanted to: “protect Google’s core business given tough competition in the computational advertising market from firms such as Facebook and Amazon;²⁹ diversify their business by entering consolidated industries led by large incumbents with conventional pipeline as opposed to platform businesses, such as telecommunications, transportation and logistics, energy, retail and healthcare, to name but a few;³⁰ and explore, discover and create businesses that may lead to the formation of entirely new industries from scratch.”³¹

The first objective did not require a corporate restructuring. As a public company, this objective was not only aligned with the interests of shareholders. Google also had the organizational alignment needed to execute a strategy of innovation ecosystem extension in order to protect its core business. An example of this strategy was the extension of Google’s innovation ecosystem with retailers such as Costco, Target and Walmart to compete with its archrival Amazon. Launched in March 2018 and branded as Shopping Actions, Google provided these new stakeholders with the best available search services to improve the product search experience of end consumers from different devices, thereby increasing order intake for its retail partners. The strategy also consisted in supercharging the output logistics of these new stakeholders with Google Express, a service of Google for order fulfillment.³²

The second objective consisted in the aggressive diversification of Google’s core business through new services in new industries. Although Google had already explored the diversification of its business prior to the creation of Alphabet, it was clear that this objective was not completely aligned with the short-term financial objectives of shareholders and that any agenda of diversification around these “Other Bets” would lack organizational alignment under the old corporate structure. To solve this problem, Google executed what can be regarded today as the most comprehensive strategy of innovation ecosystem extension.

Google founded Alphabet on October 2, 2015 and spun out many of its original investment projects in unrelated industries as independent firms under Alphabet. According to CBSInsights,³³ as of August of 2018 the firms acquired or spun out by Google that are now under Alphabet operate in areas as diverse as the life sciences, urban planning, venture capital and private equity, autonomous vehicles, broadband connectivity, Internet security, and transportation and logistics.

²⁸ See report at: <https://www.cbinsights.com/research/report/google-strategy-teardown/>.

²⁹ These are business areas referred to as “Alpha Bets” by the founders.

³⁰ These are business areas referred to as “Other Bets” by the founders.

³¹ These are projects referred to as the “moonshots” by the founders due to their speculative nature, which make them subject to deep strategic uncertainty.

³² See <https://www.cbinsights.com/research/report/google-strategy-teardown/>.

³³ See report at: <https://www.cbinsights.com/research/report/google-strategy-teardown/>.

The projects Google has been developing to achieve the third objective are subject to radical uncertainty (Kay and King 2020) and focus on radical innovations. These so-called “moonshot” projects are incubated under “X, the Moonshot Factory,”³⁴ a firm founded in 2010 that is part of the “Other Bets.” Under the new corporate structure, Alphabet provides these “Other Bets” with the flexibility needed to enable the kind of entrepreneurial management required to drive this ambitious innovation ecosystem extension strategy under uncertainty.

Google’s new grand strategy gives rise to a trade-off between granting interested parties and stakeholders high independence and the resources needed to execute different strategies, on the one hand, and retaining control, on the other. This strategy also provides a common vision, complementary resources, capabilities and competencies, and the shared rewards that are needed to organizationally align all interested parties and stakeholders under Alphabet as the holding company.

As opposed to Twitter’s strategy, Google’s strategy is more ambitious and retains a higher level of orchestration of the extended innovation ecosystem based upon leveraging economies of scope across the different stakeholders involved. Chief among them is the leadership of Google in the development of the general-purpose technology platforms that will enable the Industry 4.0, and the IoT as its new transportation infrastructure, such as Artificial Intelligence and quantum computing.³⁵

4 Discussion

The cases analyzed in Section 3 provide strong evidence suggesting that digital platforms often need to reject the internalization hypothesis of the standard model (Teece 1986, 2006, 2018a, b) in order to resolve the dilemma of innovation ecosystem extension introduced in Section 2.6. We begin this section by analyzing an alternative model of how digital platforms profit from innovation that departs from the standard model by rejecting the internalization hypothesis.

4.1 Toward a model of how digital platforms profit from innovation

By dropping the internalization hypothesis digital platforms owners can let interested parties explore, discover, develop and deploy not only resources but also capabilities and competencies with the aim of exploring multiple innovation pathways simultaneously. This begs the question of how digital platforms can entice interested parties to develop the cospecialized assets that are required to unlock the value of big data and profit from innovation in external environments that are subject to deep and even radical strategic uncertainty (Kay and King 2020). The cases analyzed in Section 3 suggest that there is indeed a logic behind the magic of innovation ecosystem extension.

³⁴ See <https://x.company/>.

³⁵ See <https://www.cbinsights.com/research/report/google-strategy-teardown/>.

4.1.1 The logic behind the magic of innovation ecosystem extension

The formation of an extended innovation ecosystem by a dominant digital platform follows a process that unfolds along the following lines:

1. global dematerialization leads to big data and does not require that incumbent digital platforms deploy resources in the countries in which their users reside, which is one of the reasons why the process of global expansion unfolds much more straightforwardly for digital platforms than for the “brick-and-mortar” businesses of the old economy;³⁶
2. digital platforms are left with large volumes of unutilized big data, whose potential value needs to be unlocked by exploring and discovering new uses (value creation) and new revenue models to monetize them (value capture);
3. digital platforms that position big data as a generalized complementary asset can accelerate and expand innovations in time and space by extending their innovation ecosystems and enticing interested parties and their stakeholders to be an integral part of them;
4. interested parties and their stakeholders deploy their own resources to explore and discover the cospecialized assets required to find new uses for the big data provided to them by incumbent digital platforms as a generalized complementary asset with the overarching goal of value creation and value capture, thereby positioning themselves as drivers of innovation;³⁷
5. as new offerings are discovered and delivered to new “stakeholders,” new viable revenue models for value capture are explored, discovered and deployed, and the new rents generated are shared in a spirit of “coopetition” following a principle of optimizing utility for the extended innovation ecosystem;
6. in exceptional cases digital platforms may internalize these assets after communicating well in advance their decision to do so to all interested parties involved based on best governance practices (Gawer and Cusumano 2002).

Note that the rents generated and shared between interested parties, stakeholders and the incumbent digital platforms in the spirit of “coopetition” mentioned above constitute a new type of rents that goes beyond the Schumpeterian rents of neo-Schumpeterian frameworks (Paredes-Frigolett and Pyka 2022). They are based on finding a balanced distribution of rents between interested parties, stakeholders, and the incumbent digital platforms. Such balanced distribution entails an optimization problem that can pursue different goals, such as maximizing the value delivered to the extended innovation ecosystem, as in the case of Google versus Apple discussed in Section 3. However, the solution to such optimization problem will generally exclude the maximization of rents for the platforms alone, which would amount to fully adopting ex-ante the internalization hypothesis, as in the case of Myspace analyzed in Section 3.

³⁶ Deployed through the Internet, the users of digital platforms can access their applications anywhere and anytime through different devices without the need for digital platforms to have a local presence. So-called “brick and mortar” businesses, however, need to deploy local resources in the places in which they operate.

³⁷ Note that digital platforms may attempt to find these new uses and viable revenue models in stand-alone mode, but as we have seen in Section 3, this strategy of internalizing the needed cospecialized assets will result in fewer innovations and will ultimately undermine the competitive advantage of digital platforms.

4.1.2 Rejecting the internalization hypothesis

The process described above allows platforms to explore several innovation pathways simultaneously under deep strategic uncertainty with a focus on value creation during exploration and value capture during exploitation. This strategy of ecosystem extension is based on rejecting the internalization hypothesis and allows platforms to unlock the value hidden in big data by deploying resources and capabilities not owned by them but rather contributed to them as “cospecialized assets” by an increasingly larger number of interested parties.

Digital platforms that fully embrace the internalization hypothesis may attempt to explore, discover and develop the needed cospecialized assets themselves without engaging interested parties, but then the processes of exploration and discovery leading to value creation and value capture would be limited to the internal resources, capabilities, and competencies owned and controlled by the digital platforms without recourse to external sources. Such strategy compromises the success of digital platforms, especially under deep strategic uncertainty (Kay and King 2020), as the case of MySpace analyzed in Section 3.2.1 shows. Even digital platforms with vast resources and a strong cash position often have to abandon the internalization hypothesis, as the case of Alphabet analyzed in Section 3.2.5 shows. The case of Alphabet analyzed in Section 3.2.6 is also an excellent example of how an incumbent digital platform extended its innovation ecosystem by spinning out a number of “intrapreneurial agendas,” endowing their spin-offs with all the resources, capabilities, and competencies needed to drive their strategies independently while at the same time retaining a fair degree of orchestration and control.

4.2 The rise of the global stakeholder capitalism model of digital platforms

Companies operating under the global hypercompetition model (D’Aveni 1994) and the global stakeholder capitalism model (Hsieh 2004; Nolan 2008) prior to the inception of the Internet ran so-called “brick-and-mortar” businesses in linear value chains (Parker et al. 2016; van Alstyne et al. 2016). After the inception of the Internet for commercial applications in the mid 1990s, many of these companies transformed their business models into “click-and-mortar” business models.³⁸ While they embraced the Internet, initially for customer-facing functions and later on for back-office operations, these business models retained the legacy of the old pipeline business models.

A limited form of dematerialization could take place through the use of information and communication technologies prior to the emergence of the Internet. Normann (2001) used the related term “liquification” in the 1980s to refer to the process needed to augment the density of offerings of companies through information and communication technologies. However, the extent to which this was possible prior to the adoption of Internet technologies for commercial applications was very limited. While incumbents that embraced Internet business models in several industries took dema-

³⁸ While the value driven by many click-and-mortar pipeline businesses led to supply-side economies of scale and was governed by Metcalfe’s law (Hendler and Golbeck 2008), the value driven by platform businesses led to demand-side economies of scale and was governed by the law of the pack (Reed 2001).

terialization a step further, they still retained the logic of pipeline business models. Although some of them deployed personalization and recommendation systems based on Internet segmentation strategies such as micro- and autosegmentation,³⁹ the degree of dematerialization these incumbents could achieve and the scale economies they could capitalize upon were limited by network effects that did not grow exponentially with the number of users.

The same applies to the first era of Internet applications that mushroomed after 1995, often referred to as Web 1.0 applications, which operated under the logic of pipeline business models whose competitive advantage was based on owning and controlling the access to the best content as a VRIN⁴⁰ resource. As they were not in the business of enabling key interactions among multiple sides following the logic of a digital platform business, as defined in Section 2, the degree of dematerialization achievable through them was still very limited.

Although digital platforms can capitalize on stronger network effects and demand-side economies of scale, achieving dematerialization on a global scale is not easy. Digital platforms need to reach and maintain a critical mass of active users across different regions to capitalize upon global dematerialization. As the few digital platforms that achieve this goal tend to take all or almost all the multisided markets they serve, incumbent digital platforms are not numerous.

But digital platforms can expand and take all or almost all the market in a much more straightforward manner than conventional pipeline businesses once they become incumbent. Key factors contributing to this are: (i) the strength of the network effects digital platforms are able to generate, which lead to demand-side economies of scale that allow digital platforms to drive hypergrowth; (ii) the access to the Internet as a public infrastructure allowing digital platforms to offer their services anywhere and anytime without the need to deploy vast resources in the countries in which their users are located; (iii) the financial backing of tier-1 venture capital firms to support digital platforms during the initial phases of their lifecycle, including the launch-to-market and expansion phases;⁴¹ (iv) the lack of frameworks regulating the Internet both globally and locally in the jurisdictions in which digital platforms operate and their users reside; and last but not least (v) the fact that existing antitrust legislation has not yet been amended to effectively limit the expansion of digital platforms globally, for example by regulating them as public utilities (Khan 2017).

4.3 Global dematerialization and the decoupling of economics activities

The process of global dematerialization is based on the human-computer interactions that revolve around the value unit of digital platforms. Such interactions take place between the multiple sides served by digital platforms, which can be located anywhere, and the applications that are made available by digital platforms and are deployed using cloud computing technologies. Since digital platforms, as defined in Section 2, draw

³⁹ Autosegmentation is based on offering a great variety of products in the long tail (Anderson 2006).

⁴⁰ VRIN stands for valuable, rare, inimitable and nonsubstitutable.

⁴¹ This support plays a crucial role, especially for digital platforms that struggle with the problem of finding a viable business model.

upon user-generated content in group-forming networks in which users are positioned not only as consumers but also as producers,⁴² the positioning of user interactions as economic activities becomes self-evident. Although we can conclude that digital platforms do contribute to the dispersion of economics activities, the deployment of information and communication technologies and the process of global dematerialization driven by digital platforms do not necessarily lead to a decoupling of economics activities in general. In fact, the opposite can be stated if we consider the evolution of digital technologies since the 1950s.

Digital and Internet technologies are not the same as the cloud computing technologies that have given rise to digital platforms. Digital technologies commenced to be commercially deployed in the 1950s and 1960s, initially following the in-house delivery paradigm, which required that corporations maintained large information systems departments to develop and maintain software applications in different functional areas. Advances in distributed computing throughout the 1980s and 1990s allowed firms to distribute information systems across regions through multi-tier client-server architectures. Two milestones in the process of adopting digital technologies are worth mentioning.

The first milestone was the transition from information systems developed in-house by large firms in several industry sectors to so-called productized enterprise software solutions. This transition gave rise to the enterprise software industry, which was pioneered by companies such as JD Edwards and SAP and consolidated as a result of the Y2K problem at the turn of the millennium. But the licensing delivery model adopted by the new enterprise software industry presupposed that corporations not only own the required software licenses, typically under licensing and yearly maintenance contracts, but also the hardware infrastructure, which was owned and managed in-house. The inception of the Internet in the mid 1990s prompted the emergence of new systems architectures allowing a greater degree of decoupling of some economic activities, mostly those involving customer-facing functions, but they also resulted in a concentration of other economic activities.

Enabled by the rapid adoption of Internet technologies, the second milestone consisted in the replacement of enterprise software solutions commercially deployed using a licensing model with enterprise software solutions deployed using new delivery models such as the managed services and the outsourced delivery models. This development had as a direct effect the transition of management tasks and functions from corporations dispersed anywhere in the world to the few companies that pioneered this next transition in the enterprise software industry. This transition to managed services and outsourced delivery models concentrated economic activities around the software vendors that led the enterprise software industry.

More recently, the consolidation of cloud computing has taken this development to the next level with the software-as-a-service (SaaS) delivery model, which was pioneered by companies such as Salesforce.com during the first decade of the new millennium. The adoption of cloud computing as enabling distributed computing technology has given rise to the infrastructure-as-a-service (IaaS) and platform-as-

⁴² They are thus referred to as “prosumers” in the digital platforms literature.

a-service (PaaS) delivery models, which have contributed to a further concentration of economic activities.⁴³

While digital platforms can in many cases lead to a decoupling of economic activities from their original geographical locations, they also contribute to the concentration of new economic activities. The process of global dematerialization plays a central role in the creation of such new activities. A case in point here was the creation of the online ad syndication market analyzed in Section 3, which ignited Google's first period of hypergrowth that led to its successful IPO in 2004. Other examples include the drop-shopping program of Amazon, which provided order fulfillment services to small retailers distributed across different geographical areas.

But digital platforms can, and in all likelihood will continue to, contribute to a further concentration of economic activities as well. In fact, they can take the creation and concentration of economic activities to the next level as orchestrators of the extended innovation ecosystems of the Industry 4.0.

4.4 The dual role of big data

Big data is a core asset of digital platforms that is valuable and rare. To the extent that the applications offered by digital platforms enable interactions that are unique, we can assume that big data is also inimitable. The uniqueness of the interactions driven by digital platforms also suggests that big data is nonsubstitutable. But the key question is not whether big data is a VRIN asset of digital platforms but rather whether it can have a dual role as a core and also as generalized complementary asset that can be offered to interested parties. The adoption of such a dual role amounts to rejecting the internalization hypothesis, as discussed in Section 4.1.2. Although digital platforms can position big data only as a core asset by embracing the internalization hypothesis, our analysis reveals that this dual role has important strategic implications for digital platforms.

First of all, this dual positioning of big data is one of the key elements that differentiates digital platform strategy from the strategy of conventional (pipeline) businesses, including the internal and external platforms described in Section 2. Another element is the fact that the structural barriers to entry that digital platforms can build are fleeting, as strong negative network effects can destroy an incumbent digital platform suddenly and often unexpectedly, as the case of Netflix shows (Wingfield and Stelter 2011). As a result, the strategy of digital platforms abandons the exclusive focus on either owning VRIN resources, capabilities and competencies (Barney 1991; Penrose 1995) or on building structural barriers to entry (Porter 2008) and follows a new strategic management approach whose focus is on stakeholder engagement as the source of a temporary competitive advantage (D'Aveni 1994).

But the most important strategic implication of this dual role of big data lies in a new area that has so far remained unexplored by scholarly research, namely, the potential to deploy metadynamic capabilities to spark process of exploration using quantum man-

⁴³ As noted by Storper and Leamer (2001), these developments in distributed computing based on the Internet as a network of networks have not only contributed to a "deagglomeration" but also to an "agglomeration" of economic activities.

agement approaches (Paredes-Frigolett 2023). The superpositioning effect mentioned in Section 1.2 and exemplified by some of the cases analyzed in Section 3, especially the cases describing the catch-up strategy of Google and its new grand strategy under Alphabet, is one of the quantum management effects that digital platforms can capitalize upon to drive the rapid exploration and discovery of new uses for increasingly larger volumes of big data efficiently and effectively under deep strategic uncertainty (Kay and King 2020).

4.5 Stakeholder engagement as a source of competitive advantage

As mentioned, digital platforms intermediate interactions around a so-called “value unit,” defined as the unit of value around which the interactions that flow through the platform revolve (Parker et al. 2016).⁴⁴ The interactions that flow through the platform involve multiple sides, including different types of users. From the perspective of the multiple sides involved, be they consumer, producers or “prosumers,” the interactions take place locally through zero-client applications that run on their devices. But from the perspective of the digital platform, these interactions take place anywhere and anytime and therefore can occur globally, which allow digital platforms to personalize their applications unobtrusively and on the fly using machine learning and user modeling based on user profiles that are built on a global scale (Rich 1979; Kobsa 2001; Schafer et al. 2007). The personalization of the interactions achieved on such a global scale reinforces the sense of locality perceived by the multiple sides involved.

Digital platforms do not need to deploy resources locally to achieve global dematerialization. This allows them to grow according to market dynamics that are different from those of firms that operate conventional, that is, pipeline business models (van Alstyne et al. 2016).⁴⁵ Stakeholder engagement is thus an important component of digital platform governance. While principles of good governance lead to stakeholder engagement (Cusumano and Gawer 2002; Gawer and Cusumano 2014; Cusumano et al. 2019), stakeholder engagement leads to more interactions flowing through the platform. As more interactions flow through the platform more data-harvesting opportunities arise. This leads to an increase in global dematerialization, which in turn leads to more stakeholder engagement by rejecting the internalization hypothesis with the aim of unlocking the value hidden in big data, driving innovation, and profiting from it in environments subject to deep strategic uncertainty (Kay and King 2020). This virtuous circle shows that global dematerialization, big data, and stakeholder engagement go hand in hand and allow digital platforms to deploy resources, capabilities and competencies they do not need to own and control.

⁴⁴ For example keywords for Google or online videos for YouTube.

⁴⁵ Digital platforms whose “value units” correspond to hard goods are a notable exception to this, as their business models retain the legacy of supply-chain platforms, which forces them to deploy input and output logistics locally as an essential component of their business model.

5 Conclusions

In this concluding section, we discuss current challenges and future directions of the model presented in Section 4. We begin this section by analyzing current threats to the business models of digital platforms.

5.1 Can the business of digital platforms be sustained?

Global dematerialization is one of the key assumptions of the global stakeholder capitalism model of digital platforms and is also the key assumption of the model of how digital platforms drive and profit from innovation.

A first threat to global dematerialization is the threat of changing the regulatory status quo.

5.1.1 The threat of regulation

Unlike pipeline businesses operating as monopolies in regulated markets (Parker et al. 2016; van Alstyne et al. 2016), digital platforms have been criticized because the lack of Internet regulation and antitrust legislation has so far allowed them to operate as global firms that exploit significant market power without being subject to the constraints that are usually imposed upon regulated monopolies (Khan 2017, 2018). As a result, digital platforms are currently under attack due to their potential abuse of monopolistic power.

Contrary to conventional monopolies and oligopolies exerting significant market power, global digital platforms do not necessarily pose an immediate threat to their clients or users, who rather benefit from innovations in the form of a higher density of free or low-priced services enabled by revenue models that draw upon near-zero marginal costs.⁴⁶ The real problem rather lies in the new market limitations, including new negative externalities, arising in the extended innovation ecosystems of digital platforms.⁴⁷

Political communications, lobbying as well as preemptive strategies of self-regulation and self-restrain may not be sufficient to keep the threat of regulation and digital tax and antitrust legislation at bay. Digital platforms will be able to provide bespoke services to governments based on global dematerialization in return for favorable regulation (or lack thereof). But they will also need to negotiate bespoke tax regimes and other forms of compensation to avoid or preempt the introduction of new digital and antitrust legislation.⁴⁸

A second threat to global dematerialization and the business model of digital platforms is posed by blockchain technologies.

⁴⁶ Although end consumers may in some cases end up being negatively affected through price discrimination (Clemons 2019).

⁴⁷ As the rise in property rentals as a result of the business model of Airbnb exemplifies (Nieuwland and van Malik 2020).

⁴⁸ Such as the digital taxation introduced by Airbnb (<https://www.airbnb.com/help/article/2283/occupancy-tax-collection-and-remittance-by-airbnb-in-canada>). Antitrust legislation has also been proposed with the aim of regulating digital platforms as public utilities (Khan 2017, 2018).

5.1.2 The threat of blockchain technologies

Glocal, as opposed to global dematerialization, refers to a new dematerialization paradigm where there is no single centralized point controlling the harvesting of big data and the access to and control over it. Glocal dematerialization is predicated upon having multiple peers exercise such command and control locally over subsets of data. Distributed globally throughout the network, these multiple peers can grant access to big data using a fully federated services model. This model is local in terms of data harvesting and control yet global in that the network of peers expands globally and can grant access to data and federate services anywhere and anytime. Blockchain technologies thus have the potential to replace global dematerialization with glocal dematerialization.

Today, incumbent digital platforms do not embrace the decentralized and fully federated model of the Internet proposed by blockchain technologies. They use a centralized model of global dematerialization instead. Although the case can be made that blockchain technologies do not yet scale to implement mission-critical systems in several applications domains, the reasons for maintaining a centralized model are ultimately not technical. A model based on the fully federated approach envisioned by blockchain technologies could, in principle, be developed and thrive. This begs the question of whether the services provided by digital platforms will be deployed using such a fully federated model in the future.

As neither the digital platforms themselves nor other key interested parties and stakeholders in their extended innovation ecosystems will relinquish ownership of and control over the process of global dematerialization, the most likely scenario will be the adoption of blockchain technologies not to dismantle but rather to further consolidate the global stakeholder capitalism model of digital platforms, as set out by Paredes-Frigolett and Pyka (2022). Will such scenario of further consolidation be accelerated by technological developments such as the IoT and the IoE⁴⁹ as the new transportation infrastructure of the Industry 4.0?

Firms in key industry verticals that are today becoming early adopters of data federation through the IoT are excellent candidates to adopt blockchain technologies and become interested parties in the future extended innovation ecosystems of digital platforms. The integration of blockchain technologies with the IoT and the IoE will result in a dramatic expansion of these ecosystems and in novel forms of dematerialization that will not only need to be explored, discovered, and negotiated between the platforms and new interested parties and stakeholders but also exploited and monetized by them. The IoT and the IoE will thus require a great deal of federation of data and services. The integration of blockchain technologies will help digital platforms, and the interested parties and stakeholders in their extended innovation ecosystems, federate increasingly larger volumes of data and services on a large global scale. This begs the question of who will play the key role in this process of federation of data and services in the future.

As orchestrators of the extended innovation ecosystems arising in connection with the IoT and the IoE, incumbent digital platforms are in a privileged position to maintain

⁴⁹ The Internet of Things and the Internet of Everything.

global dematerialization as the key political assumption of the global stakeholder capitalism model of digital platforms and to keep their role as orchestrators of their extended innovation ecosystems. There are several reasons why digital platforms may even be able to expand this orchestrating role based on a new semi-federated model that will allow digital platforms to take advantage of blockchain technologies without relinquishing the control they currently exert over big data and ultimately over global dematerialization.

Chief among these reasons is the fact that key stakeholders in their extended innovation ecosystems, such as government agencies at national level and transgovernmental organizations at supranational level, may have a vested interest in letting dominant digital platforms keep their role of orchestrators not only of their extended innovation ecosystems but also of the data-industrial complexes that will arise around them. The fact that the new business models that will arise in connection with the IoT, the IoE and the Industry 4.0 will be platform as opposed to pipeline business models (Parker et al. 2016) is also a factor that may contribute to the further consolidation of the role of dominant digital platforms as orchestrators of their extended innovation ecosystems.

We anticipate that the most likely scenario is for blockchain technologies, the IoT and the IoE to augment the power of global dematerialization and the amount of big data generated and monetized in the extended innovation ecosystems of digital platforms. Although this scenario will present a myriad of new opportunities for digital platforms and their interested parties and stakeholders, it will also render the extended innovation ecosystems of digital platforms highly complex. As these new extended innovation ecosystems will be ridden with ethical dilemmas, they will be more difficult to govern. We therefore anticipate that a new form of extended governance will be of high strategic importance to sustain the global stakeholder capitalism model of digital platforms in the future and that a new stakeholder, namely, civil society, will play a key role in the extended governance of digital platforms.

5.1.3 The threat of civil society, online ad-hocracies, and online mobs

Ethical issues arising as a result of interactions not only with public-sector organizations and governments but also with users directly can lead to unexpected responses from incumbent digital platforms as well as from civil society, online ad-hocracies, and online mobs. Such responses can drive negative network effects that can grow exponentially in a short period of time and can threaten the business model of digital platforms, giving competitors the opportunity to replace them. Digital platforms are more exposed to this threat than conventional pipeline businesses, as their own users and civil society can mobilize large online ad-hocracies and online mobs and spark processes of value destruction that can undermine an otherwise viable business model in a short period of time.⁵⁰

Recognizing the potential of civil society organizations to become a mechanism of governance in the future, we anticipate a scenario in which digital platforms will need to increase the delivery of human and public goods in society jointly with civil

⁵⁰ The case of Netflix losing 800,000 active accounts, and the goodwill of its users, in a few days in 2011 shows us how serious this threat can be (Wingfield and Stelter 2011).

society organizations and other key stakeholders to mitigate or compensate for negative externalities affecting the multiple sides they serve. Digital platforms will probably be well advised to do so not for reputational effects but rather in the pursuit of a genuine goal of serving society.⁵¹

5.2 Digital platforms and the Industry 4.0

Platforms are currently entering and disrupting mature industries such as retail, transportation and logistics, energy, education, and healthcare. This begs the question of whether the dominant digital platforms or the current incumbents in these industries will play the orchestrating role in the future.

One could argue that incumbents owning and controlling VRIN resources, including tangible assets that are difficult to imitate and substitute, would be better positioned to assume the orchestrating role in the extended innovation ecosystems that will arise in connection with current developments such as the IoT, the IoE, and the Industry 4.0. Despite the clear threats to the business models of digital platforms, some of which we addressed in Section 5.1, we would argue that digital platforms are in principle better positioned to assume this orchestrating role for several reasons.

Digital platforms exert control over end consumer data across industries, which gives them an informational advantage over incumbent firms with conventional business models. Dominant digital platforms have also the financial muscle and the flexibility needed to execute grand strategies based on a wide range of dynamic capabilities (Eisenhardt and Martin 2000; Teece 2007; Paredes-Frigolett 2023).⁵² Dominant digital platforms have also assumed a position of leadership when it comes to the development of general-purpose technology platforms that will propel the Industry 4.0, such as Artificial Intelligence and cloud computing, and they are also well positioned to lead the transition from classical to quantum computing. Digital platforms have also vast experience in the implementation of data-driven business models, which are poised to become the business models that will be adopted by the IoT, the IoE and the Industry 4.0.

Another argument that may be posited regarding the advantages of digital platforms to assume the orchestrating role in the extended innovation ecosystems that will arise in connection with the Industry 4.0 vis-à-vis incumbent firms with conventional business models is an empirical fact that has been described by digital platform scholars, namely, that “it is easier to add product functionality to a platform than adding platform functionality to a product” (Parker et al. 2016). We also anticipate that dominant digital platforms with a strong cash position will embark in a series of strategic acquisitions of incumbents with conventional business models as part of their pathways to disruption in several industries.

For all these reasons, we anticipate that digital platforms will be better positioned to orchestrate the extended innovation ecosystems that will emerge in connection with

⁵¹ Such as green platforms to support the transformation toward higher degrees of sustainability (Pyka and Stöber 2023).

⁵² The grand strategy of Google under Alphabet analyzed in Section 3 is a good example of the implementation of such dynamic capabilities.

the Industry 4.0.⁵³ But that being said, digital platforms remain fragile and vulnerable not only for the reasons outlined in Section 5.1 but also because their business models are inherently subject to the so-called general strategic problem of digital platforms.

5.3 The general strategic problem of digital platforms

Departing from the internalization hypothesis, digital platforms extend their innovation ecosystems by deploying big data as a complementary asset with the aim of accelerating and expanding innovations in time and space. Central to such strategies of innovation ecosystem extension is the process of global dematerialization, which creates the need to unlock and harness the value hidden in big data. As already mentioned, one of the main challenges global digital platforms are currently facing is the contingent threat that new Internet regulation and legislation, including antitrust legislation, pose to the sustainability of global dematerialization. The main strategy of digital platforms to meet this challenge has so far consisted in avoiding this threat via self-regulation and preemption.

Another challenge lies in the very nature of the network effects that digital platforms can draw upon to drive demand-side economies of scale, gain critical mass, and consolidate as incumbent platforms. While positive networks effects can propel the hypergrowth of digital platforms in a very short period of time based on value creation in group-forming networks, negative network effects⁵⁴ can destroy value and cause the demise of incumbent digital platforms just as rapidly, especially in the absence of tangible assets that may protect digital platforms from being dethroned as incumbent platforms by latent competitors.

Our analysis suggests that digital platforms will be well advised to embrace new models of extended governance not only to render tighter regulatory regimes less appealing in society but also to keep the threat posed by negative network effects at bay. Digital platforms shall do so by executing strategies aimed at increasing the positive externalities as well as the distribution of human and public goods in society not for narrow interests but with a genuine dual role of treating all stakeholders as ends in themselves. Although the above-mentioned threats already show how fragile the business models of digital platforms are, they only scratch the surface of a much more serious threat to the sustainability of their business models, which can be referred to as “the general strategic problem of digital platforms.”

The general strategic problem of digital platforms consists in solving the complex ethical “multilemmas” that will arise in the extended innovation ecosystems of digital platforms.⁵⁵ We argue that this problem will render digital platforms more fragile due to the sheer complexity of their future extended innovation ecosystems. To address this problem, we anticipate that digital platforms will need to deploy new frameworks for

⁵³ Google, after the creation of Alphabet, is showing us how the execution of its grand strategy is leading to the formation of such extended innovation ecosystems (<https://www.nytimes.com/2015/08/11/technology/giving-google-room-to-dream-big-beyond-search.html>).

⁵⁴ Often driven by their own sides, as in the case of Netflix (Wingfield and Stelter 2011).

⁵⁵ The general strategic problem of digital platforms is an extension of the general problem of the firm proposed by Singer (2013) for pipeline businesses in which the value creation process takes place in a linear value chain.

the extended governance of their extended innovation ecosystems encompassing not only the usual corporate strategy and business ethics dimensions but also the political and international relations dimensions as well.

Key assumptions of such frameworks shall include: (i) the adoption of a dual mission in society, that is, one that recognizes a multiple fiduciary duty of digital platforms to serve and treat all stakeholders in their future extended innovation ecosystems, including civil society, as “ends in themselves and not purely instrumentally” (Singer 2013, p. 316); (ii) the engagement in political communications and lobbying aimed at upholding and extending the negative corporate freedoms of digital platforms, allowing them to continue to drive innovations in their extended innovation ecosystems jointly with organizations from the public and private sector; and (iii) the recognition of the moderating effect that digital platforms can have on international relations, geopolitics and geostrategy in the future due to their privileged position as orchestrators of the data-industrial complexes arising around them.

Our future work will focus on the development and implementation of formal frameworks for the extended governance of digital platforms aimed at solving the general strategic problem of digital platforms based on decision-theoretic approaches.

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Data Availability The authors confirm that all the data sources supporting the findings of this study are referenced in this article and were publicly available at the time of submission.

Declarations

Conflicts of interest The authors declare that they have no conflict of interest.

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